



Archetypal Graph Generative Models

Explainable & Identifiable Communities via Anchor-Dominant Convex Hulls

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1 Motivation & Contributions

Graph embeddings give strong link-prediction and clustering scores, but the latent coordinates carry symmetry ambiguity and explanations are added after training. GraphHull turns the latent geometry itself into the explanation.

i

Hierarchical design

Two-level convex hulls: K global archetypes + K anchor-dominant local hulls.

ii

Identifiability

Truncated Dirichlet with $\varepsilon < \frac{1}{2}$ $\Rightarrow$ disjoint hulls $\Rightarrow$ unique reconstruction.

iii

Stable & scalable

Boxed-SVD $\Rightarrow$ Lipschitz gradient. Edge subsampling: $O(N^2) \Rightarrow O(|E|)$.

iv

Principled diversity

DPP priors on global and local archetypes prevent collapse and degeneracy.

v

Empirical validation

Competitive or superior on link prediction & community detection.

2 Two-Level Convex Hull Model

K **global archetypes** A define extremal community profiles. For each k , a **local prototypical hull** B_k refines that community with K vertices: 1 fixed anchor a_k and $K-1$ prototypes.

$$A = [a_1^\top; \dots; a_K^\top] \in \mathbb{R}^{K \times D}$$

ANCHOR CONSTRUCTION (eq. 1)

$$B_k = \widetilde{W}_k A = \begin{bmatrix} W_k \\ e_k^\top \end{bmatrix} A \in \mathbb{R}^{K \times D}$$

Bottom row of B_k = anchor a_k . Other $K-1$ rows are convex combinations of global archetypes.

$$(W_k)_{\ell,:} \in \Delta_{K-1} := \{w \in \mathbb{R}^K : w \geq 0, \mathbf{1}^\top w = 1\}, \quad \ell = 1, \dots, K-1$$

$$\omega_i \sim \text{Dir}(\alpha_\omega \mathbf{1}_K), \quad c_i \sim \text{Cat}(\pi)$$

$$z_i = \omega_i^\top B_{c_i} \in \mathbb{R}^D$$

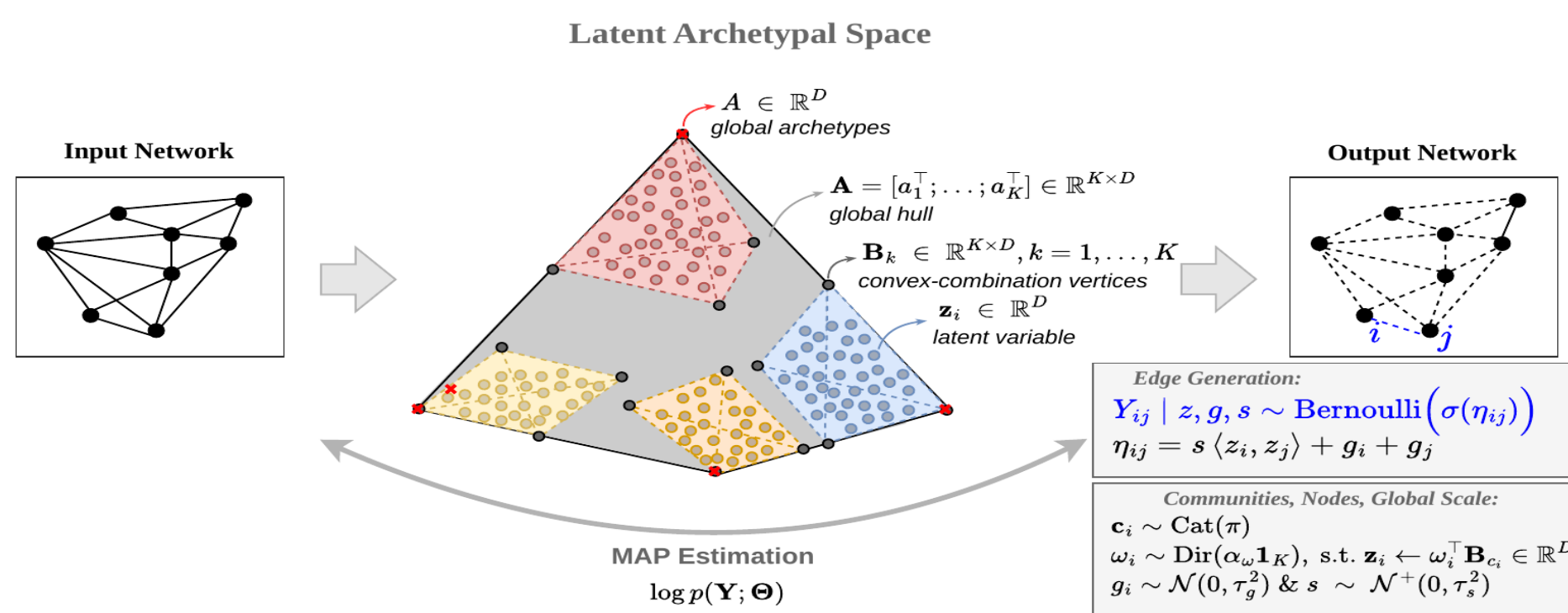


Figure 1. Two levels of convex hulls. Global archetypes A (red) define the outer hull; each local hull B_k captures one community.

WHAT THIS GIVES

Per node: explicit (c_i, ω_i) — the embedding z_i is fully determined by them.

Per community:

a hull anchored at one global archetype with $K-1$ prototypes.

Per prediction:

the edge logit splits into geometric inner product + degree biases.

3 Identifiability via Anchor Dominance

We need disjoint local hulls so every node embedding admits a unique (community, prototype) decomposition. We obtain disjointness BY CONSTRUCTION via a truncated Dirichlet prior parametrized by an anchor mass ε .

$$w \sim \text{Dir}(\alpha \mathbf{1}_K) \mid w_k \geq 1 - \varepsilon, \quad 0 < \varepsilon < \frac{1}{2}$$

ANCHOR-DOMINANT IMPLEMENTATION (eq. 7)

$$(W_k)_{r,:} = (1 - s_{k,r}) e_k^\top + s_{k,r} \tilde{q}_{k,r}, \quad s_{k,r} = \varepsilon t_{k,r}$$

$\varepsilon \in (0, \frac{1}{2})$ controls anchor dominance. Smaller $\varepsilon \Rightarrow$ tighter, fully separated hulls.

$$\tilde{q}_{k,r} \sim \text{Dir}(\alpha_q \mathbf{1}_K), \quad t_{k,r} \sim \text{Beta}(a, b)$$

LEMMA 1 — Non-overlap of anchor-dominant local hulls

$$\varepsilon < \frac{1}{2} \implies \text{conv}(B_k) \cap \text{conv}(B_\ell) = \emptyset \quad (k \neq \ell)$$

$\Rightarrow$ every $z_i \in \text{conv}(B_{c_i})$ has a unique community-level reconstruction.

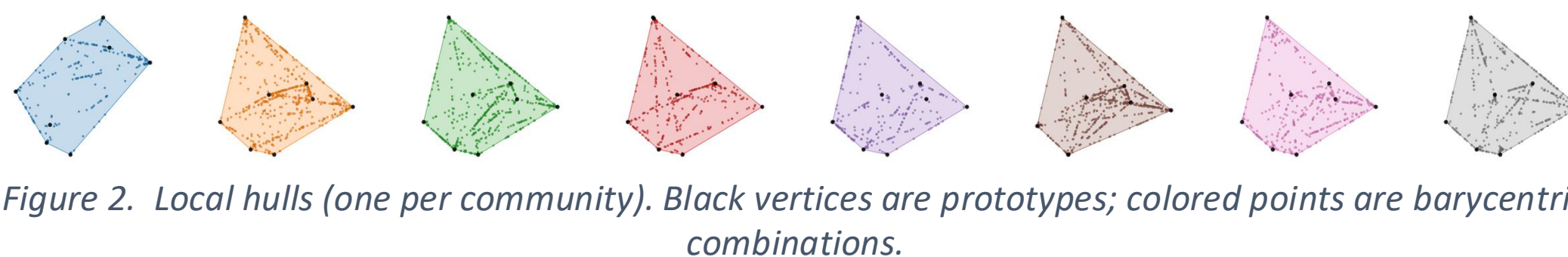


Figure 2. Local hulls (one per community). Black vertices are prototypes; colored points are barycentric combinations.

4 Generative Process

1 Global archetypes:

$$A \in \mathbb{R}^{K \times D}$$

2 Local hulls:

$$B_k = \widetilde{W}_k A$$

3 Diversity priors on:

$$A, B_k$$

4 Community + node:

$$c_i \sim \text{Cat}(\pi), \quad \omega_i \sim \text{Dir}(\alpha_\omega \mathbf{1}), \quad z_i = \omega_i^\top B_{c_i}, \quad g_i \sim \mathcal{N}(0, \tau_g^2)$$

5 Global scale:

$$s \sim \mathcal{N}^+(0, \tau_s^2)$$

6 Edges:

$$Y_{ij} \sim \text{Bernoulli}(\sigma(s \langle z_i, z_j \rangle + g_i + g_j))$$

WHY THIS WORKS

Identifiability: anchors are unique vertices of each hull, so disjointness follows by construction (Lemma 1).

Expressivity:

ε tunes hull size while keeping disjointness — only $\varepsilon < \frac{1}{2}$ is required.

Scalability:

observed-edge term + non-edge subsampling $\Rightarrow$ per-iteration cost $O(|E|)$.

5 Edge Likelihood & MAP Objective

Binary edges via Bernoulli likelihood with logit equal to a global-scale inner product plus degree biases.

$$\eta_{ij} = s \langle z_i, z_j \rangle + g_i + g_j$$

$$Y_{ij} \mid \mathbf{Z}, \mathbf{g}, s \sim \text{Bernoulli}(\sigma(\eta_{ij}))$$

$$\log p(\mathbf{Y} \mid \mathbf{Z}, \mathbf{g}, s) = \sum_{i < j} [Y_{ij} \eta_{ij} - \log(1 + \exp \eta_{ij})]$$

JOINT MAP OBJECTIVE (eq. 12)

$$\log p(\mathbf{Y}, \Theta) = \sum_{(i,j) \in \mathcal{D}} [Y_{ij} \eta_{ij} - \text{softplus}(\eta_{ij})]$$

$$+ \sum_{k=1}^K [\log \det(\mathbf{L}_k) - \log \det(\mathbf{I} + \mathbf{L}_k)]$$

$$+ [\log \det(\mathbf{L}) - \log \det(\mathbf{I} + \mathbf{L})]$$

$$- \frac{1}{2\tau_g^2} \sum_{i=1}^N g_i^2 - \frac{s^2}{2\tau_s^2}$$

Bernoulli likelihood + DPP priors on local & global archetypes + Gaussian/half-normal priors on biases and scale.

6 Boxed SVD & Lipschitz Stability

$$A = U \text{diag}(\sigma) V^\top, \quad \sigma_j \in [\sigma_{\min}, \sigma_{\max}]$$

$$\|A\|_2 \leq \sigma_* \implies \|z_i\| \leq \kappa_*$$

THEOREM 1 — Boxing $\Rightarrow$ Lipschitz gradient in ω

$$L_\omega \leq \frac{1}{4} \left[s \kappa_*^2 \deg_{\max} + \frac{s^2}{2} \kappa_*^4 \deg_{\max} \right]$$

Projected gradient with $\eta \leq 1/L_\omega$ is provably safe — monotone descent.

7 Diversity & Scalability

$$\log p_{\text{DPP}}(\Phi) = \log \det(\mathbf{L}) - \log \det(\mathbf{I} + \mathbf{L}), \quad \mathbf{L} = \Psi \Psi^\top$$

DPP priors act as a repulsive force on local prototypes B_k and global archetypes A — preventing degenerate or co-linear vertices.

Per-iteration cost ($K \ll N$):

$$\mathcal{O}(|E|) + \mathcal{O}(K^3 D + K^4)$$

PRACTICAL RECIPE

Set σ_* : spectral box bounds the embeddings; start from the empirical scale of A .

Set $\varepsilon \in (0, \frac{1}{2})$:

smaller $\varepsilon \Rightarrow$ tighter, fully separated hulls; larger $\varepsilon \Rightarrow$ more flexibility.

GS-relaxed c_i :

differentiable assignments at training; hard one-hot at inference.

8 Interpretability & Self-Explainability

TRANSPARENCY

Archetypes & prototypes appear directly in the generative mechanism — no post-hoc explainer required. Every coefficient in the model has a geometric meaning.

DIVERSITY

DPP priors on global and local archetypes + affine independence of A + anchor-dominant non-overlap by construction. Hulls cover diverse, expressive volumes.

TRUSTWORTHINESS

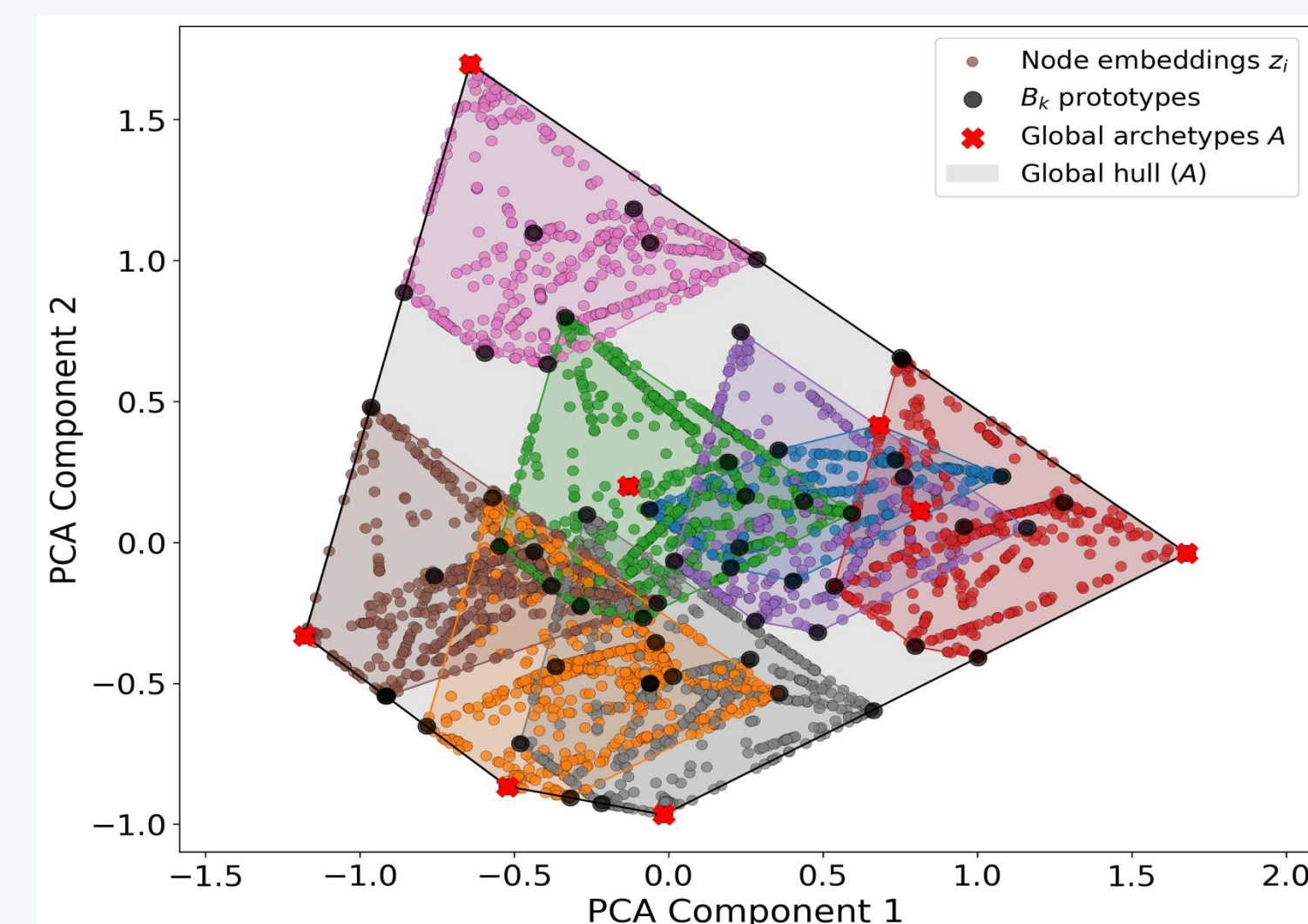
Faithful generative decomposition + boxed-SVD spectral box $\Rightarrow$ stable optimization with controlled curvature. Theorem 1 gives a Lipschitz gradient bound.

GENERATIVE CONSISTENCY

The same interpretable factors (c_i, ω_i, B_k, A) that explain a prediction also generate the data. Geometry IS the explanation.

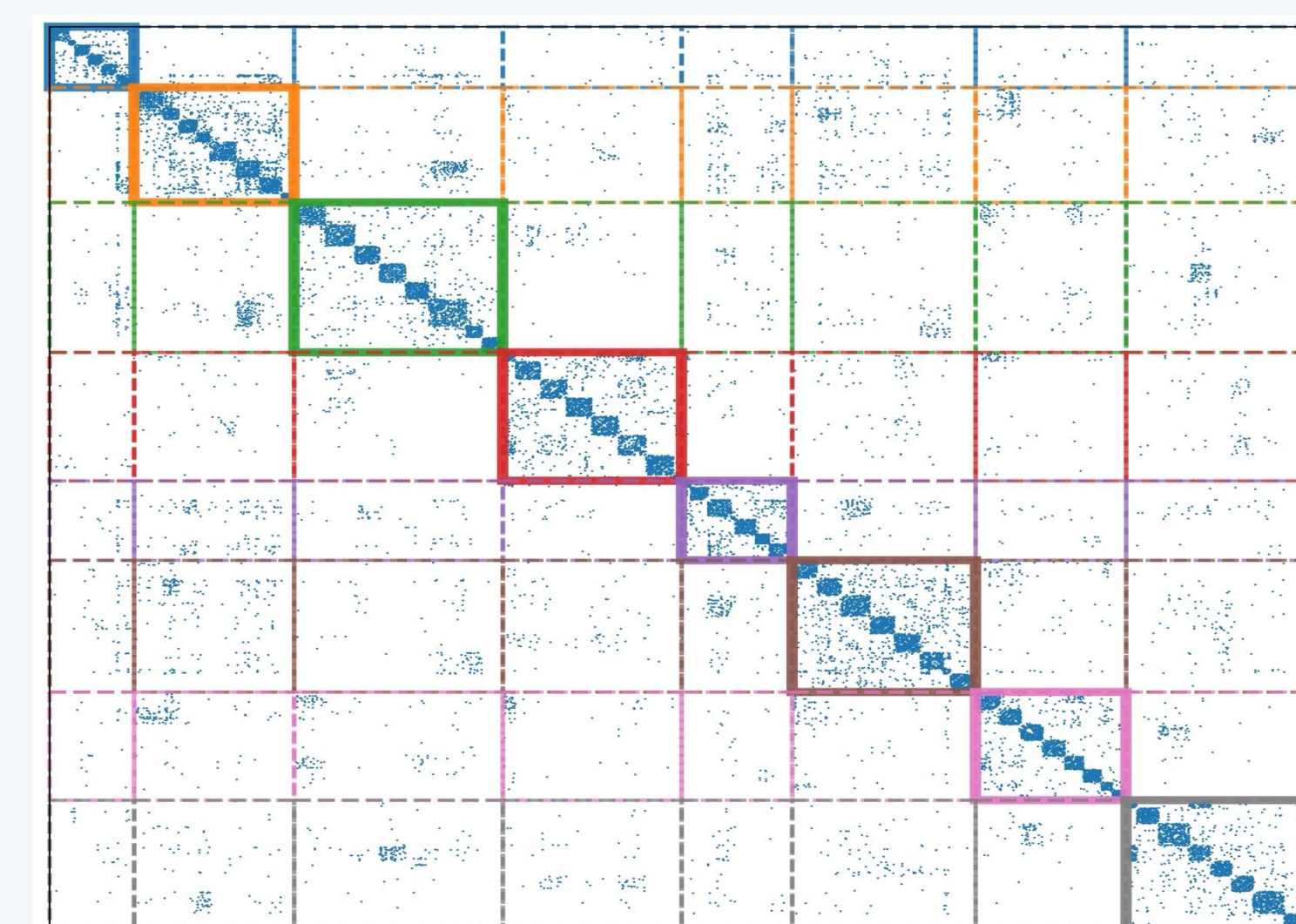
9 Results — What the model recovers

Latent geometry (PCA)



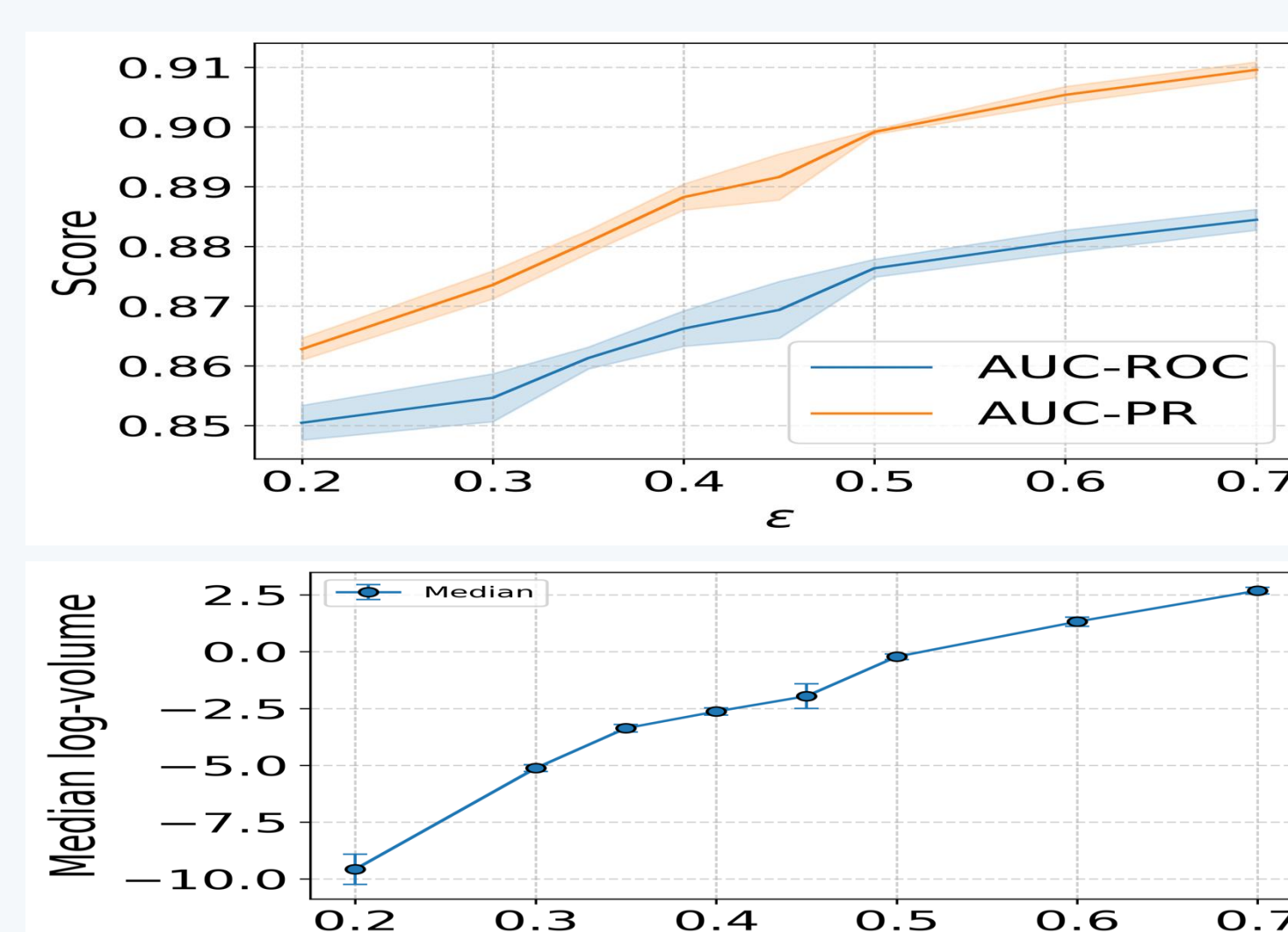
Local hulls disjoint; global archetypes form the outer envelope.

Reordered adjacency



Reordered first by global hull assignments, then within blocks by $\argmax \omega_i$.

Effect of ε on performance



Top: AUC vs ε (HepTh). Bottom: median log-volume of local hulls.

Link prediction (AUC-ROC) — D=64

Method	AstroPh	GrQc	HepTh	Cora	DBLP
Node2Vec	.962	.936	.892	.777	.941
Role2Vec	.965	.934	.895	.752	.944
MNMF	.954	.918	.875	.680	.931
Dmon	.861	.866	.807	.788	.791
GraphHull	.965	.945	.904	.803	.948

Node clustering (NMI / ARI)

Dataset	Cora	Citeseer	LastFM	Pol
Best NMI	.486	.320	.605	.727
GraphHull NMI	.422	.347	.616	.757
Best ARI	.417	.320	.522	.791
GraphHull ARI	.333	.372	.514	.820

TAKEAWAY

GraphHull is identifiable, interpretable, and competitive — geometry IS the explanation. Anchor-dominant local hulls give unique node decomposition; boxed-SVD makes optimization safe; DPP priors keep the hulls expressive.